

Egor Serebriakov

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EDUCATION

ITMO University M.Sc. Student in Artificial Intelligence, GPA 4.00	St. Petersburg, Russia (Remote) Sep 2025 - Present
Brown University M.Sc. in Physics	Providence, RI Aug 2022 - May 2024
Moscow State University B.Sc. in Physics with Honors	Moscow, Russia Sep 2018 - Jun 2022

RESEARCH EXPERIENCE

Research Collaborator, Mila CERC-AAI Lab · Advisor: Prof. Irina Rish	Montréal, QC Mar 2026 - Present
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- Designed and implemented layerwise probing, effective-rank, and cross-dataset transfer experiments across multiple datasets and model backbones to study specialization in time-series foundation models.
- Studying orchestration architectures for autonomous LLM agents and AI-scientist systems, focusing on self-evaluation, monitoring, and reliable execution under constrained compute budgets.

Research Assistant, ITMO University iMak AI Lab · Advisor: Dr. Andrei Zakharov	St. Petersburg, Russia Sep 2025 - Feb 2026
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- Investigated RL-based methods to adaptively control causal weighting in Physics-Informed Neural Network (PINN) training, inspired by *"Respecting Causality Is All You Need"*, for strongly nonlinear and chaotic dynamical systems.
- Demonstrated on the Allen-Cahn equation that a single adaptive RL-driven training run can match the performance of fixed causal parameters selected via multiple training sweeps.

Research Assistant, Brown University AI for High-Energy Physics · Advisor: Prof. Loukas Gouskos	Providence, RI Jan 2024 - Jun 2024
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- Investigated knowledge distillation for particle jet classification by transferring information from a large-scale graph neural network teacher model to student models trained in two settings: with reduced input resolution and with simplified architecture.
- Achieved >3x inference speedup through teacher-student distillation while preserving classification performance (AUC ≈ 0.9), enabling faster jet tagging with the ultimate goal of developing models suitable for real-time deployment at CERN.

Undergraduate Researcher, Russian Quantum Center Magnetoplasmonics and Ultrafast Magnetism · Advisor: Prof. Vladimir Belotelov	Moscow, Russia Dec 2020 - Jun 2022
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- Conducted femtosecond pump-probe magneto-optical experiments on ultrafast spin dynamics and compared measured resonance frequencies with theoretical models of magnetization dynamics.

MANUSCRIPT IN PREPARATION

Probing Representation Tunnels in Time-Series Foundation Models: Specialization, Geometry, and Transfer

E. D. Serebriakov*, Z. Shang*, A. Mitra, and I. Rish (*Equal contribution)

Manuscript in preparation, 2026

We investigate whether representation tunnels emerge in time-series foundation models by measuring forecasting performance, effective rank, and cross-dataset transfer across layers before and after fine-tuning. We study how late-layer specialization and representational compression relate to generalization under distribution shift.

AWARDS & HONORS

Sep 2025	Best EdTech Project, <i>AI Talent Hackathon 2025</i>	Online
Jun 2024	Winner, <i>Junior ML Contest</i>	St. Petersburg, Russia
Jun 2022	Diploma with Honors (Top 6%), <i>Faculty of Physics, Moscow State University</i>	Moscow, Russia
Sep 2021	Academic Excellence Scholarship, <i>Moscow State University</i>	Moscow, Russia

SELECTED PRESENTATIONS

Jan 2026	RL-Controlled Physics-Informed Neural Networks, <i>AI Talent Conf 2026</i>	St. Petersburg, Russia
Jun 2025	AI for High-Energy Physics Experiments, <i>Junior ML Contest</i>	St. Petersburg, Russia
May 2024	Accelerating Inference in Jet Tagging via Distillation, <i>DL Day at Brown</i>	Providence, RI

SELECTED PROJECTS

Automated Video Editing Pipeline

ASR-Driven Video Segmentation and Assembly · Python, FFmpeg, faster-whisper, Adobe Premiere Pro	Jul 2026
- Built an end-to-end video system: GPU word-level ASR (Whisper large-v3), source-relative timestamp alignment, schema-validated LLM clip selection, and frame-exact assembly on a shared integer-frame timeline. - Made the pipeline safely resumable with validated intermediate artifacts and comprehensive unit tests, and automated the Adobe Premiere Pro round trip using FCP7 XML, media relinking, and remapped captions.	

Education Platform – AI Talent Hackathon 2025

EdTech Category Winner · GitHub	Sep 2025
Built a web-based education platform integrating an LLM for automated grading and an interactive student assistant.	

Anomaly Detection at the Large Hadron Collider (CERN)

Graduate Course Project, Brown University	Jan - May 2024
- Developed convolutional and attention-based autoencoders for unsupervised anomaly detection in LHC jet data. - Evaluated the models using synthetic anomaly injections and tested transfer across datasets.	

ADDITIONAL EXPERIENCE

Student, CERN Computational & Data Science Summer School

Princeton University · Princeton, NJ	Jul 2024
- Completed training in parallel programming, optimization, and collaborative software development. - Studied applications of data science and machine learning in particle physics research.	

Teaching Assistant

Brown University · Providence, RI	Sep 2022 - May 2024
Teaching Assistant for four physics courses: mentored students during lab sessions, graded coursework, and led problem-solving workshops.	
Beloretsk STEM Center · Beloretsk, Russia	2016 - 2018
Instructed high school students in mathematics and physics over two semesters.	

SKILLS & INTERESTS

Research Interests: Large Language Models, Time-Series Foundation Models, Representation Learning, Generalization and Transfer, and AI Agents

Programming: Python (PyTorch, NumPy, pandas, scikit-learn), C

Tools & Compute: Git, Linux, Bash, GPU Training, HPC Clusters

Work Authorization in Canada: Open Work Permit